Test Data

$z_1 = \begin{pmatrix} 30 \\ 15 \end{pmatrix}$ y=60 $z_2 = \begin{pmatrix} 40 \\ 20 \end{pmatrix}$ y=80 $z_3 = \begin{pmatrix} 50 \\ 25 \end{pmatrix}$ y=100

Outlier X_s	NO Regularizer with (as if λ=0)	Ridge Regularizer with λ=3	Ridge Regularizer with λ=10
11 3 22	W = (0, 2, 0) ||W||²=4 Test₁: x₁=30, x₂=15, y=60 Test₂: x₁=40, x₂=20, y=80 Test₃: x₁=50, x₂=25, y=100	W = (0.1, 1.95, 0.03) ||W||²=3.8 Test₁: x₁=30, x₂=15, y=59 Test₂: x₁=40, x₂=20, y=78.7 Test₃: x₁=50, x₂=25, y=98.4	W = (0.13, 1.87, 0.09) ||W||²=3.5 Test₁: x₁=30, x₂=15, y=57.4 Test₂: x₁=40, x₂=20, y=76.5 Test₃: x₁=50, x₂=25, y=95.6
30 3 22	W = (2.7, 0.67, 0.64) ||W||²=8.11 Test₁: x₁=30, x₂=15, y=32.4 Test₂: x₁=40, x₂=20, y=42.3 Test₃: x₁=50, x₂=25, y=52.3	W = (1.2, 0.72, 0.77) ||W||²=2.45 Test₁: x₁=30, x₂=15, y=34.1 Test₂: x₁=40, x₂=20, y=45.1 Test₃: x₁=50, x₂=25, y=56.1	W = (0.54, 0.73, 0.81) ||W||²=1.48 Test₁: x₁=30, x₂=15, y=34.6 Test₂: x₁=40, x₂=20, y=46 Test₃: x₁=50, x₂=25, y=57.4
60 3 22	W = (4, 0.29, 0.78) ||W||²=16.7 Test₁: x₁=30, x₂=15, y=24.4 Test₂: x₁=40, x₂=20, y=31.2 Test₃: x₁=50, x₂=25, y=38	W = (1.77, 0.32, 0.98) ||W||²=4.2 Test₁: x₁=30, x₂=15, y=26.2 Test₂: x₁=40, x₂=20, y=34.4 Test₃: x₁=50, x₂=25, y=42.5	W = (0.82, 0.33, 1.1) ||W||²=1.92 Test₁: x₁=30, x₂=15, y=26.8 Test₂: x₁=40, x₂=20, y=35.4 Test₃: x₁=50, x₂=25, y=44.1